

SESSION 3: LATENCY OPTIMIZATION STRATEGIES

The ultimate goal: making our models fast enough for real-world deployment.



Memory Bandwidth Optimization

2. In-place Operations:

Core message

Avoid unnecessary copies; they double memory traffic and pressure.

Many ops (ReLU, clamp, simple norms, elementwise arithmetic) can overwrite inputs safely.

Benefits

Halves bandwidth for the op; reduces peak memory footprint.

Especially valuable on MCUs and also pays off on desktops by reducing DRAM traffic.

Safeguards

Only do in-place when no later step needs the original tensor.

Watch for aliasing with views/slices.



Component Power Breakdown:

SESSION 2: ENERGY-EFFICIENT PARALLEL PROCESSING

**Power Consumption Models –
Understanding Energy Usage Across
Scales**

1. Data Center Scale:

- CPU: 500-2000mW
- GPU: 250W (training), 75W (inference)
- Memory: 5-10W per DIMM
- Network: 10-50W per high-speed port

2. Mobile/Edge Scale:

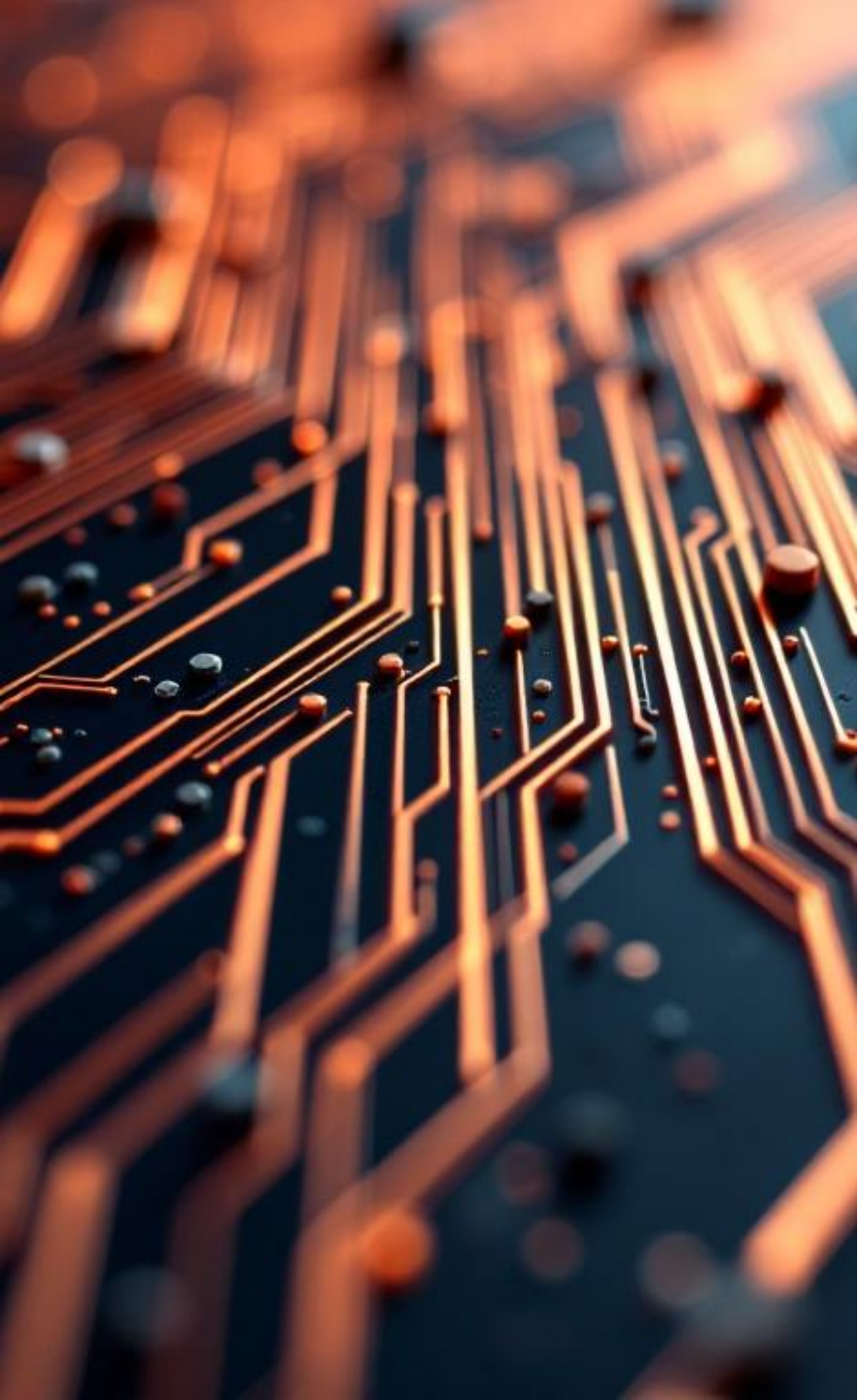
- Application processor: 1-5W
- Display: 400mW
- Radio (active): 800mW
- Memory: 100-500mW

3. Microcontroller Scale:

MCU: 1mW (Cortex-M4F at 48MHz)
Sensors: 1mW (accelerometer),
300 μ W (microphone)
Radio: 40mW (BLE), 100mW (WiFi)
Memory: Negligible for SRAM

Key Insight:

Radio communication dominates power at all scales.



Energy-Power Relationship

Fundamental Energy Concepts

Power vs. Energy:

- Power = Energy / Time (watts = joules/second)
- For battery life: Average power determines duration
- 1mW average → 2x longer life than 2mW average

Battery Capacity Examples:

- CR2032 coin battery: 2,500J (1 month at 1mW)
- AA battery: 15,000J (6 months at 1mW)
- Phone battery: ~50,000J

Energy Harvesting Potential:

- Indoor light: $10\mu\text{W}/\text{cm}^2$
- Outdoor light: $10\text{mW}/\text{cm}^2$
- Industrial thermal: $1\text{-}10\text{mW}/\text{cm}^2$
- Vibration: $10\text{-}100\mu\text{W}/\text{cm}^2$

Design Implication:

Only outdoor solar and industrial thermal currently practical for self-powered ML devices.

Parallel Processing Power Models

Energy Implications of Parallelism

Sequential vs. Parallel Power:

Sequential Execution:

- Single core: 1mW
- Execution time: 100ms
- Energy: 0.1mJ

Parallel Execution (4 cores):

- Four cores: $4 \times 1\text{mW} = 4\text{mW}$
- Execution time: 25ms
- Energy: 0.1mJ (same total energy)

Power-Performance Trade-offs:

- Parallelism reduces latency without changing total energy
- May enable more aggressive duty cycling
- Additional hardware increases baseline power
- Must consider parallel efficiency losses

SIMD Power Benefits:

Single instruction, multiple data operations:

- Process 4 values with 1.2× power of single value
- 3.3× energy efficiency improvement
- Available on ARM Cortex-M4 and higher

Energy Efficiency in AI Hardware

Power Consumption Across the Computing Stack

Energy Cost Hierarchy:

- 32-bit floating-point addition: ~20 femtojoules
- On-chip SRAM access: ~5 picojoules
- Off-chip DRAM access: ~1.3 nanojoules (65,000x more than computation!)

femto-: 10^{-15} (one quadrillionth)

pico-: 10^{-12} (one trillionth)

nano-: 10^{-9} (one billionth)

Optimization Strategies:

1. Minimize Data Movement

- Maximize on-chip computation before external memory access
- Reuse data across multiple operations

2. Precision Reduction

- 8-bit integer operations consume ~5x less energy than 32-bit floating point
- Specialized low-precision arithmetic units

3. Specialized Circuits

- Fixed-function units more energy-efficient than programmable logic
- Elimination of instruction fetch and decode overhead

4. Voltage and Frequency Scaling

- Lower voltage operation for energy-constrained environments
- Dynamic frequency scaling based on workload requirements

Energy-Aware Design Optimization

Minimizing Power Consumption

Power sources to watch

1. Computation energy

- 8-bit multiply $\approx$ 20 femtojoules
- 16-bit multiply $\approx$ 80 femtojoules
- 32-bit multiply $\approx$ 320 femtojoules
- Memory read/write: roughly 5–1000 picojoules depending on the level

2. Memory access energy

- Register access is cheapest
- Cache access costs a few to tens of picojoules
- DRAM access is most expensive: 1–10 nanojoules

3. Peripheral energy

- Sensor sampling: microjoules per sample
- Radio transmission: millijoules per packet

Optimization strategies

Reduce computation

- Prefer bit shifts and adds over multiplies when mathematically equivalent
- Use lower precision (int8/int16) where accuracy allows
- Fuse operations and reuse intermediate results

Minimize memory access

- Process data in place to avoid copies
- Keep hot data in registers or on-chip SRAM; avoid DRAM
- Arrange data for cache-friendly, sequential access
- Use fixed-size buffers and avoid dynamic allocation

Duty cycling

- Sleep between capture and inference cycles
- Batch work: wake briefly, process, then return to deep sleep
- Keep radios off; transmit only summaries or alerts at low duty cycle